

Part 1. Data Orientation

What social determinants of health are, how Healthy People 2030 organizes them, which 8 unique datasets this toolkit uses across 10 entries, and how to get started.

Audience: Undergraduate and MPH students | **Level:** Beginner | **Time:** 1 hour

Learning Objectives

- Define social determinants of health (SDOH) using Healthy People 2030 (HP 2030).
- Identify the five HP 2030 SDOH domains and provide at least two examples of variables for each.
- Name the 8 datasets (10 table entries) in this toolkit and match each dataset to its primary domain.
- Explain what county **Federal Information Processing Standards (FIPS)** codes are and why all datasets use them for merging.
- Describe the three open scholarship requirements: cite, reproduce, and share.

1. Social Determinants of Health (SDOH) as Research Variables

Health is not just about genes, personal choices, or access to doctors. Research consistently shows that where people are born, grow up, live, work, and age shapes their health outcomes more than medical care alone. These conditions are called social determinants of health, or SDOH.

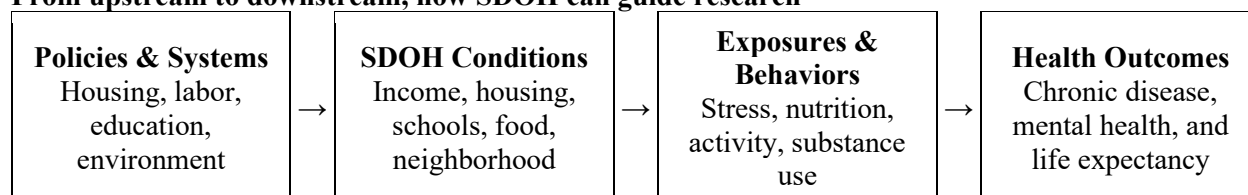
“Social determinants of health are the conditions in the environments where people are born, live, learn, work, play, worship, and age that affect a wide range of health, functioning, and quality-of-life outcomes and risks.”

[Healthy People 2030, U.S. Department of Health and Human Services](#)

SDOH are useful for open scholarship because they connect real-world social conditions to measurable indicators that can often be examined using publicly available data. Depending on the research question, indicators such as income, housing, education, food access, or neighborhood conditions may be used as independent variables, control variables, or outcomes of interest.

A useful way to think about this is that a person's health is influenced far upstream, long before they ever see a doctor. A child growing up in a neighborhood with high poverty, poor schools, food deserts, and environmental pollution faces a fundamentally different health trajectory than one who does not regardless of individual behavior.

From upstream to downstream, how SDOH can guide research



This toolkit focuses on exploring the SDOH conditions using publicly available data. By understanding where SDOH burdens are concentrated, researchers and students can identify communities most in need of policy intervention and support evidence-based decision-making.

2. The Five Healthy People 2030 SDOH Domains

Healthy People 2030 organizes SDOH into five domains. Every dataset in this toolkit maps to one or more of these domains. Understanding the domains helps you choose the right data for your research question.

SDOH Domain 1: Economic Stability

- Focuses on whether people have sufficient economic resources to maintain health, including income, employment, food security, housing stability, and financial strain.
- Example variables include poverty rate, unemployment rate, median household income, SNAP receipt, and housing cost burden.



SDOH Domain 2: Education Access & Quality

- Focuses on educational opportunity, school quality, early childhood education, literacy, and educational attainment.
- Example variables include high school graduation rate, educational attainment, school enrollment, chronic absenteeism, and limited English proficiency.

Domain 3: Health Care Access & Quality

- Focuses on insurance coverage, primary care access, provider shortages, preventive services, quality of care, and barriers to care, especially in rural and resource-limited areas.
- Example variables include uninsured rate, primary care shortage areas, cancer screening rates, preventable hospitalizations, and mental health providers.

SDOH Domain 4: Neighborhood & Built Environment

- Focuses on housing quality, food access, environmental pollution, transportation, walkability, parks, climate risk, and physical conditions of where people live.
- Example variables include food desert status, environmental burden index, housing cost burden, air quality/PM2.5, and disaster risk.

SDOH Domain 5: Social & Community Context

- Focuses on social support networks, civic participation, incarceration, community safety, social cohesion, and family and community structure.
- Example variables include violent crime rate, social associations, and voter participation

Note: These five domains do not operate separately. A community with high poverty often also has poor schools, fewer doctors, more pollution, and higher social vulnerability. These conditions tend to cluster geographically, which is why multi-domain analysis, combining data across all five domains, may be more powerful than studying any single domain alone. The datasets in this toolkit are designed to be merged and analyzed together.

3. The 10 Dataset Entries (8 Distinct Sources)

This toolkit utilizes 10 dataset entries from 8 sources. The American Community Survey (ACS) covers both Economic Stability and Education Access & Quality Domains. All sources are publicly available. Access methods vary by source, and some API-based workflows require a free API key. Geographic identifiers such as FIPS codes are standardized as needed before merging data in R.

What is the FIPS (Federal Information Processing Standards) code?

- A FIPS code is a 5-digit number that uniquely identifies U.S. counties. The first 2 digits represent the state, and the last 3 digits the county. For example, the FIPS code for Sangamon County, Illinois, is 17167. Here, 17 is Illinois, and 167 is Sangamon County.
- All datasets in this toolkit use FIPS codes to merge data in R.
- Store FIPS as a string, not a number, to keep leading zeros like Alabama's 01. Storing as a number removes zeros, causing data merge issues.

Domain 1 — Economic Stability Datasets

Dataset	What it measures	Geography	Format	Download URL
American Community Survey (ACS 5-Year) U.S. Census Bureau	Poverty, income, unemployment, SNAP receipt, housing cost burden, education, health insurance, disability, and language proficiency	Census Tract, Zip code tabulation area (ZCTA), County, State, National	CSV (Comma-Separated Values) files, API, FTP	Census Table: Income, Poverty & Employment
County Health Rankings & Roadmaps (CHR&R) Univ. of Wisconsin Population Health Institute	County-level health factors and outcomes, including unemployment, income inequality, child poverty, food insecurity, social associations, and violent crime	County, State, National	CSV, Excel, GitHub	National data & documentation

Domain 2 — Education Access & Quality Datasets

Dataset	What it measures	Geography	Format	Download URL
ACS 5-Year — Education variables U.S. Census Bureau	Educational attainment, school enrollment, preschool enrollment, limited English proficiency, and related household/social indicators	Census Tract, ZCTA , County, State, National	CSV, API, FTP	Census Table: Education
NCES Common Core of Data (CCD) National Center for Education Statistics	Public school and district characteristics, enrollment, free/reduced-price lunch eligibility, teacher/staff counts, school type, charter status, and Title I status	School, District, State, National	CSV	CCD Data Files

Domain 3 — Health Care Access & Quality Datasets

Dataset	What it measures	Geography	Format	Download URL
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<u>CDC PLACES: Local Data for Better Health</u> Centers for Disease Control and Prevention	Measurements include local estimates of health outcomes, behaviors, status, prevention practices, and non-medical factors from the ACS.	Census Tract, ZCTA, County Place	CSV, API	<u>PLACES Data Download</u> <u>PLACES Data Portal</u>
<u>HRSA Health Professional Shortage Areas</u> Health Resources & Services Administration	Primary care, dental, and mental health shortage designations; HPSA score, status, type, and rural status	Service Address, County, State, National	CSV	<u>HRSA Data Warehouse</u>

Domain 4 — Neighborhood & Built Environment Datasets

Dataset	What it measures	Geography	Format	Download URL
<u>ATSDR Environmental Justice Index (EJI)</u> Agency for Toxic Substances and Disease Registry	Air quality, water quality, proximity to hazardous sites, climate burden, social vulnerability, health vulnerability — combined into composite EJI score	Census Tract, County	CSV, Geodatabase	<u>EJI Data Download</u>
<u>USDA Food Access Research Atlas</u> U.S. Department of Agriculture	Low-income and low-access food tracts, distance to nearest supermarket, share of households without vehicle access to food, food desert classification	Census Tract, County	CSV, Excel	<u>Food Access Research Atlas – Download the Data</u>

Domain 5 — Social & Community Context Datasets

Dataset	What it measures	Geography	Format	Download URL
<u>ATSDR Social Vulnerability Index (SVI)</u> Agency for Toxic Substances and Disease Registry	Overall vulnerability including (1) socioeconomic status, (2) Household Characteristics, (3) Racial & Ethnic Minority Status, (4) Housing Type & Transportation	Census Tract, ZCTA, County	CSV, Geodatabase	<u>SVI Data & Documentation Download</u>
<u>CHR&R — Civic, Community and Social Support</u> Univ. of Wisconsin Population Health Institute	Civic and community indicators such as voter turnout, social associations, broadband access, local news access, public libraries, violent crime, and related social support measures	County, State, National	CSV, Excel	<u>National data & documentation</u>

4. How To Use This Toolkit

All projects using this toolkit follow a consistent five-step workflow. These steps are the same whether you're an undergraduate working on a course project or an MPH student developing a capstone. Part 2 explains Steps 2 and 3 in detail, while Part 3 focuses on Steps 4 and 5.

1

Choose an SDOH domain and write a research question

Choose one or two HP 2030 domains. Then, formulate a research question that includes a specific population, location, and outcome. For example: “*Which Illinois counties with high poverty levels also have low cancer screening rates?*”

2

Download the data (Part 2)

Use the download links in Section 3. Save each raw file to data/raw/ without modifying it. Record the source URL, data year, and access date, as you will need all three for your citation.

3

Run the R scripts (Part 2)

Scripts #1 through #5 handle data downloading and cleaning for one or two datasets each. Script #6 merges all outputs into one master county CSV. Script #7 in Part 2 runs the statistical analyses.

4

Build your Tableau dashboard (Part 3)

Open Part 3 for step-by-step instructions on building your Tableau Public dashboard in your browser. Upload your master CSV, follow the guided exercises to build your maps, and publish your dashboard to Tableau Public. No software download needed.

5

Deposit and cite (Part 3)

Share your work openly through GitHub. When appropriate, create a citable research record with Zenodo and consider an institutional or other appropriate open repository.

5. Open Scholarship Requirements

This toolkit is part of a grant-funded open scholarship project. Every analysis you produce using these datasets should follow three open science practices so your work can be reused, verified, and built upon by others. Three open scholarship requirements for every project:

1. Cite your data. Every dataset you use must be cited in your methods section. Share your work openly on GitHub and cite all datasets used in your methods section.

2. Make it reproducible. Reproducible means someone who has never seen your project could download your files and get the same results without asking you any questions. **Six elements** make this possible:

- A. **Raw data files:** Save every downloaded file to data/raw/ exactly as downloaded and never modify it. Clean the data in R and save separately to data/clean/. Never edit the original.

- B. **Download dates:** Public datasets update without notice. Record the exact date you downloaded each file, the URLs where you downloaded directly, and the data year.
- C. **R script with detailed comments:** Use # to explain what each section does and why. For example, “# Filter to Illinois only, change 'IL' for other states.”
- D. **Script run order:** Scripts must run in sequence. Someone reproducing your work needs to know how to run Script #1 before Script #5, or it will fail.
- E. **Codebook:** List every variable in your master CSV: column name, plain-language label, source dataset, data year, and units. One row is for one variable.
- F. **README file:** one document that ties it all together. Include your name, analysis description, script run order, R version, package versions, and download dates.

3. Share your work openly. The files you created in “**2. Make it reproducible**” should go into your GitHub repository. When your project is complete, create a free GitHub account at github.com and upload your project folder as a public repository. When appropriate for your project, you may create a citable research record through Zenodo and explore institutional or other open repository options. Step-by-step instructions for all three are in Part 3.

- Your repository should contain the final master CSV, R scripts, data dictionary, README file, and other shareable files needed for reproducibility. Keep original raw files locally unless redistribution is appropriate.
- If you are using this toolkit as part of a course, follow your instructor’s directions for submitting your GitHub repository URL. If you are working independently, keep the public repository URL as part of your project record and portfolio. Selected student projects may also be featured in a course or public showcase.

The FAIR Principles

The three open scholarship requirements above (Cite, Reproduce, Share) are grounded in a widely used international framework called the **FAIR principles**, first published by Wilkinson et al. (2016) in *Scientific Data*. **FAIR** stands for:

- **F — Findable.** Your data and code should be easy for others to find. This means uploading your project to a public GitHub repository with a README and registering a permanent, citable identifier (DOI) through Zenodo, so others know what it contains.
- **A — Accessible.** Once found, your data and code should be retrievable by anyone without unnecessary barriers. Publishing your dashboard on Tableau Public, posting your scripts on GitHub as a public repository, and saving data in open file formats (CSV, not Excel) all make your work accessible.
- **I — Interoperable.** Your data should work with other datasets and tools. This is exactly why this toolkit uses FIPS codes to merge datasets. FIPS is a common, standardized identifier that links a geographic dataset to another. Using standard column names, consistent formats, and open file types (CSV, R scripts) all support interoperability.
- **R — Reusable.** Your data and code should be documented well enough that others, including yourself, can understand and reuse them. A well-commented R script, a complete codebook, dataset citations with download dates, and an open license all make your work reusable.

This toolkit is based on applying the FAIR principles. Using it helps you practice FAIR open scholarship.

FAIR	How Your Toolkit Already Teaches It	Part
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Findable	Public project documentation, data dictionary, and a persistent identifier or repository record when appropriate.	Part 2 (README and data dictionary), Part 3 (GitHub, Zenodo DOI, IDEALS)
Accessible	Tableau Public dashboard, public GitHub repository, CSV open format	Part 3
Interoperable	FIPS codes for merging, standard column names, open file formats (CSV, R scripts)	Part 1 (FIPS concept) Part 2 (FIPS in all scripts)
Reusable	Commented R scripts, codebook/data dictionary, dataset citations with download dates	Part 2

Reference: Wilkinson, M. D., et al. (2016). The FAIR Guiding Principles for scientific data management and stewardship. *Scientific Data*, 3, 160018. <https://doi.org/10.1038/sdata.2016.18>

Dataset citation templates following APA 7th edition

Copy and paste these citations into the Data Sources section of your methods. Update the access date (in brackets) to the date you downloaded the data.

American Community Survey 5-Year

U.S. Census Bureau. (2026). American Community Survey 5-year estimates, 2020–2024 [Data set]. <https://data.census.gov/>

CDC PLACES 2025

Centers for Disease Control and Prevention. (2025). PLACES: Local data for better health, county data 2025 release [Data set]. <https://data.cdc.gov/resource/swc5-untb>

CDC/ATSDR Social Vulnerability Index 2022

Agency for Toxic Substances and Disease Registry. (2025). CDC/ATSDR Social Vulnerability Index 2022 [Data set]. <https://www.atsdr.cdc.gov/place-health/php/svi/index.html>

County Health Rankings & Roadmaps 2025

University of Wisconsin Population Health Institute. (2026). County Health Rankings & Roadmaps 2025 national data [Data set]. <https://www.countyhealthrankings.org/health-data/methodology-and-sources/data-documentation>

CDC/ATSDR Environmental Justice Index 2024

Centers for Disease Control and Prevention and Agency for Toxic Substances Disease Registry. (2024). Environmental Justice Index [Data set]. <https://www.atsdr.cdc.gov/place-health/php/eji/index.html>

USDA Food Access Research Atlas

U.S. Department of Agriculture, Economic Research Service. (2021). Food Access Research Atlas: 2019 data [Data set]. <https://www.ers.usda.gov/data-products/food-access-research-atlas/>

HRSA Health Professional Shortage Areas

Health Resources and Services Administration. (2024). Health Professional Shortage Areas (HPSAs) [Data set]. <https://data.hrsa.gov/data/download>

NCES Common Core of Data

National Center for Education Statistics. (2024). Common Core of Data [Data set]. <https://nces.ed.gov/ccd/ccddata.asp>

6. Part 1 Checklist

Complete all items below before moving on to Part 2. Items 1–4 are required before you start the R scripts in Part 2. Items 5–7 are best completed after you have downloaded your selected datasets.

- ☐ I can define SDOH in my own words using Healthy People 2030.
- ☐ I can name all five HP 2030 SDOH domains and give at least one example variable from each.
- ☐ I understand what FIPS codes are and why compatible geographic identifiers are important when merging datasets.
- ☐ I have reviewed the 10 dataset entries and have identified the datasets relevant to my research question.
- ☐ I have bookmarked or written down the download links for my selected datasets.
- ☐ I understand the three open scholarship requirements: cite, reproduce, and share.
- ☐ I have reviewed the five workflow steps and know what Part 2 and Part 3 cover.

If you cannot check items 1–4, return to the relevant section before starting Part 2. If you are stuck on a concept, re-read that section or consult your instructor, mentor, or the referenced learning resources. Do not move on until items 1–4 are complete.

Ready for the next step?

Part 2 covers preparing the toolkit datasets and running seven R scripts to produce a master analysis CSV and statistical analyses.

**Part 2: Download & R
Workflow →**